

Part 1

```
yahboom@yahboom-virtual-machine: ~/Desktop/Lab4/Intro_ws $  
[INFO] [1773414569.200804540] [square_client]: Goal accepted  
[INFO] [1773414569.899377258] [square_client]: Received feedback: turtle_interfaces.msg.turtlePose(head=90.0, position=array([ 5.0000000e+01, -5.1414006e-15]), dtype=float32))  
[INFO] [1773414571.649523929] [square_client]: Received feedback: turtle_interfaces.msg.turtlePose(head=180.0, position=array([ 50., 50.]), dtype=float32))  
[INFO] [1773414573.410043059] [square_client]: Received feedback: turtle_interfaces.msg.turtlePose(head=270.0, position=array([ 7.105274e-15, 5.0000000e+01]), dtype=float32))  
[INFO] [1773414575.235025515] [square_client]: Received feedback: turtle_interfaces.msg.turtlePose(head=0.0, position=array([-1.4325892e-14, 1.4210855e-14]), dtype=float32))  
[INFO] [1773414576.247471607] [square_client]: Goal succeeded! Result: turtle_interfaces.msg.turtlePose(head=0.0, position=array([-1.4325892e-14, 1.4210855e-14]), dtype=float32))  
yahboom@yahboom-virtual-machine: ~/Desktop/Lab4/Intro_ws $ source install/setup.bash  
yahboom@yahboom-virtual-machine: ~/Desktop/Lab4/Intro_ws $ ros2 run turtle_test make_square  
[INFO] [1773414871.416488807] [square_client]: Waiting for action server...  
[INFO] [1773414871.417798010] [square_client]: Sending goal request...  
[INFO] [1773414871.464851979] [square_client]: Goal accepted  
[INFO] [1773414872.250085824] [square_client]: Received feedback: turtle_interfaces.msg.turtlePose(head=90.0, position=array([150., 0.]), dtype=float32))  
[INFO] [1773414874.011552222] [square_client]: Received feedback: turtle_interfaces.msg.turtlePose(head=180.0, position=array([50., 50.]), dtype=float32))  
[INFO] [1773414875.737899213] [square_client]: Received feedback: turtle_interfaces.msg.turtlePose(head=270.0, position=array([ 0., 50.]), dtype=float32))  
[INFO] [1773414877.561828225] [square_client]: Received feedback: turtle_interfaces.msg.turtlePose(head=0.0, position=array([-9.1848509e-15, 7.1054274e-15]), dtype=float32))  
[INFO] [1773414878.576074269] [square_client]: Goal succeeded! Result: turtle_interfaces.msg.turtlePose(head=0.0, position=array([-9.1848509e-15, 7.1054274e-15]), dtype=float32))  
yahboom@yahboom-virtual-machine: ~/Desktop/Lab4/Intro_ws $ ros2 run turtle_test make_square --ros-args -p param_square_size:=200.0  
[INFO] [1773414884.456965958] [square_client]: Waiting for action server...  
[INFO] [1773414884.458374241] [square_client]: Sending goal request...  
[INFO] [1773414884.465109447] [square_client]: Goal accepted  
[INFO] [1773414885.857966424] [square_client]: Received feedback: turtle_interfaces.msg.turtlePose(head=90.0, position=array([ 2.0000000e+02, -4.188043e-14]), dtype=float32))  
[INFO] [1773414888.461305241] [square_client]: Received feedback: turtle_interfaces.msg.turtlePose(head=180.0, position=array([200., 200.]), dtype=float32))  
[INFO] [1773414891.831466568] [square_client]: Received feedback: turtle_interfaces.msg.turtlePose(head=270.0, position=array([ 5.64342e-14, 2.0000000e+02]), dtype=float32))  
[INFO] [1773414893.823022172] [square_client]: Received feedback: turtle_interfaces.msg.turtlePose(head=0.0, position=array([-2.888186e-14, 5.684342e-14]), dtype=float32))  
[INFO] [1773414894.833289683] [square_client]: Goal succeeded! Result: turtle_interfaces.msg.turtlePose(head=0.0, position=array([-2.888186e-14, 5.684342e-14]), dtype=float32))  
yahboom@yahboom-virtual-machine: ~/Desktop/Lab4/Intro_ws $
```

Part 2

```
yahboom@yahboom-virtual-machine: ~/Desktop/Lab4/intro_...  
Summary: 1 package finished [3.93s]  
yahboom@yahboom-virtual-machine:~/Desktop/Lab4/intro_ws$ source install/setup.ba  
sh  
yahboom@yahboom-virtual-machine:~/Desktop/Lab4/intro_ws$ ros2 launch python_para  
meters run_multi_launch.py  
[INFO] [launch]: All log files can be found below /home/yahboom/.ros/log/2026-03-  
14-01-43-00-980665-yahboom-virtual-machine-12197  
[INFO] [launch]: Default logging verbosity is set to INFO  
[INFO] [minimal_param_node-1]: process started with pid [12198]  
[INFO] [minimal_param_node-2]: process started with pid [12200]  
[INFO] [minimal_param_node-3]: process started with pid [12202]  
[minimal_param_node-1] [INFO] [1773423782.946226951] [group6.Hiro]: Hello Hiro!  
[minimal_param_node-1] [INFO] [1773423783.890813550] [group6.Hiro]: Hello Hiro!  
[minimal_param_node-3] [INFO] [1773423784.414183384] [group0.NA]: Hello NA!  
[minimal_param_node-1] [INFO] [1773423784.890869668] [group6.Hiro]: Hello Hiro!  
[minimal_param_node-1] [INFO] [1773423785.891734839] [group6.Hiro]: Hello Hiro!  
[minimal_param_node-3] [INFO] [1773423786.871801997] [group0.NA]: Hello NA!  
[minimal_param_node-1] [INFO] [1773423786.891899321] [group6.Hiro]: Hello Hiro!  
[minimal_param_node-1] [INFO] [1773423787.891217401] [group6.Hiro]: Hello Hiro!  
[minimal_param_node-1] [INFO] [1773423788.892131971] [group6.Hiro]: Hello Hiro!  
[minimal_param_node-3] [INFO] [1773423789.370660732] [group0.NA]: Hello NA!  
[minimal_param_node-1] [INFO] [1773423789.891323423] [group6.Hiro]: Hello Hiro!  
[minimal_param_node-1] [INFO] [1773423790.891832379] [group6.Hiro]: Hello Hiro!
```

```
yahboom@yahboom-virtual-machine: ~/Desktop/Lab4/intro_...
yahboom@yahboom-virtual-machine:~/Desktop/Lab4/intro_ws$ source install/setup.bash
yahboom@yahboom-virtual-machine:~/Desktop/Lab4/intro_ws$ ros2 launch python_parameters run_multi_launch.py parameterName:=value
[INFO] [launch]: All log files can be found below /home/yahboom/.ros/log/2026-03-14-01-45-53-337261-yahboom-virtual-machine-12339
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [minimal_param_node-1]: process started with pid [12341]
[INFO] [minimal_param_node-2]: process started with pid [12343]
[INFO] [minimal_param_node-3]: process started with pid [12345]
[minimal_param_node-1] [INFO] [1773423955.224023036] [group6.Hiro]: Hello Hiro!
[minimal_param_node-1] [INFO] [1773423956.178106089] [group6.Hiro]: Hello Hiro!
[minimal_param_node-3] [INFO] [1773423956.713735398] [group0.NA]: Hello default!
[minimal_param_node-1] [INFO] [1773423957.181361222] [group6.Hiro]: Hello Hiro!
[minimal_param_node-1] [INFO] [1773423958.178985209] [group6.Hiro]: Hello Hiro!
[minimal_param_node-3] [INFO] [1773423959.163252051] [group0.NA]: Hello default!
[minimal_param_node-1] [INFO] [1773423959.178205982] [group6.Hiro]: Hello Hiro!
[minimal_param_node-1] [INFO] [1773423960.178707647] [group6.Hiro]: Hello Hiro!
[minimal_param_node-1] [INFO] [1773423961.179013498] [group6.Hiro]: Hello Hiro!
[minimal_param_node-3] [INFO] [1773423961.663508984] [group0.NA]: Hello default!
[minimal_param_node-1] [INFO] [1773423962.178926658] [group6.Hiro]: Hello Hiro!
[minimal_param_node-1] [INFO] [1773423963.178330828] [group6.Hiro]: Hello Hiro!
[minimal_param_node-3] [INFO] [1773423964.162950767] [group0.NA]: Hello default!
[minimal_param_node-1] [INFO] [1773423964.178261089] [group6.Hiro]: Hello Hiro!
```

```
from launch import LaunchDescription
from launch_ros.actions import Node
#### add the necessary imports
from launch.actions import DeclareLaunchArgument
from launch.substitutions import LaunchConfiguration, TextSubstitution

def generate_launch_description():
    #### add the argument declaration
    name_0_arg = DeclareLaunchArgument('my_parameter',
    default_value = TextSubstitution(text='default'))
    ####

    talk1 = Node(
        package = 'python_parameters',
        namespace = 'group6',
        executable = 'minimal_param_node',
        name = 'Hiro',
        parameters = [{'my_parameter':
'Hiro'}])

    talk2 = Node(
```

```

        'Baymax'}]],
        "--log-level", "WARN"]])

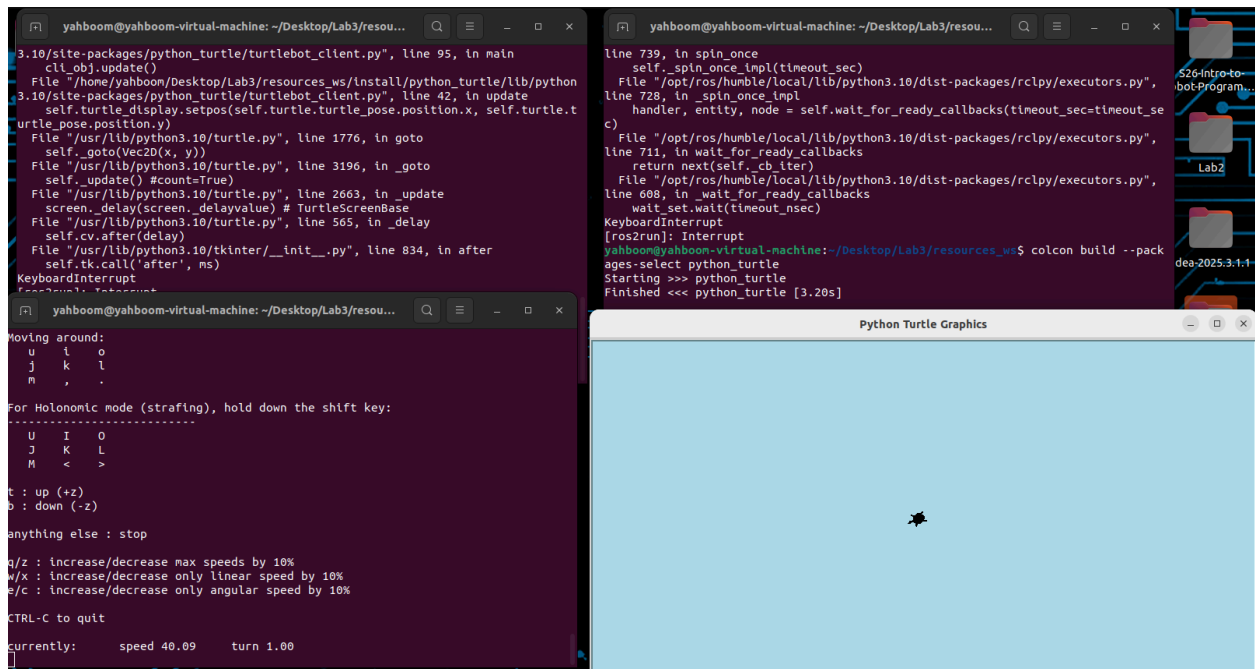
        talk3 = Node(
            package = 'python_parameters',
            namespace = 'group6',
            executable = 'minimal_param_node',
            name = 'Baymax',
            parameters = [{'my_parameter':
                'NA', 'wait_time': 2.5}]]

        LaunchConfiguration('my_parameter'), 'wait_time': 2.5}]]

    return LaunchDescription([
        name_0_arg, #argument must also be
part of the launch description
        talk1, talk2, talk3])

```

Task 1



Task 2

```
import rclpy
from rclpy.node import Node
import math
import random

import turtle

from geometry_msgs.msg import Twist, Pose

from turtle_interfaces.srv import SetColor
from turtle_interfaces.msg import TurtleMsg
from rclpy.parameter import Parameter
from rcl_interfaces.msg import ParameterDescriptor

class TurtleClient(Node):
    def __init__(self):
        super().__init__('turtleClient')

    ##### Display/Turtle Setup #####
```

```

self.screen = turtle.Screen()
self.screen.bgcolor('lightblue')
self.turtle_display = turtle.Turtle()
self.turtle_display.shape("turtle")
self.turtle = TurtleMsg()

#### publisher define ####
#self.twist_pub = self.create_publisher(Twist, 'turtleDrive', 1)
#####

self.declare_parameter('turtleColor', 'black',
ParameterDescriptor(description='turtle pen color'))
turtleColor =
self.get_parameter('turtleColor').get_parameter_value().string_value
self.turtle_display.color(turtleColor)
self.declare_parameter('turtleWidth', 1,
ParameterDescriptor(description='turtle pen width'))
turtleWidth =
self.get_parameter('turtleWidth').get_parameter_value().integer_value
self.turtle_display.width(turtleWidth)
self.color_cli = self.create_client(SetColor, 'setColor')
while not self.color_cli.wait_for_service(timeout_sec=1.0):
    self.get_logger().info('Color service not available,
waiting...')
self.color_req = SetColor.Request()
self.color_req.color =
self.get_parameter('turtleColor').get_parameter_value().string_value
self.server_call = True
self.service_future = self.color_cli.call_async(self.color_req)

#### subscribing turtlebot state ####
self.turtle_sub = self.create_subscription(TurtleMsg,
'turtleState', self.turtle_callback, 1)

def turtle_callback(self, msg):

    self.turtle = msg

def update(self):

    if self.turtle.color == 'None':
        self.turtle_display.penup()

```

```

        else:
            self.turtle_display.pencolor(self.turtle.color)

            self.turtle_display.setpos(self.turtle.turtle_pose.position.x,
self.turtle.turtle_pose.position.y)

            roll, pitch, yaw =
rpy_from_quat(self.turtle.turtle_pose.orientation.x,

self.turtle.turtle_pose.orientation.y,

self.turtle.turtle_pose.orientation.z,

self.turtle.turtle_pose.orientation.w)
            self.turtle_display.seth(math.degrees(yaw))

def quat_from_rpy(roll, pitch, yaw):

    cy = math.cos(yaw*0.5)
    sy = math.sin(yaw*0.5)
    cp = math.cos(pitch*0.5)
    sp = math.sin(pitch*0.5)
    cr = math.cos(roll*0.5)
    sr = math.sin(roll*0.5)

    qw = cr * cp * cy + sr * sp * sy
    qx = sr * cp * cy - cr * sp * sy
    qy = cr * sp * cy + sr * cp * sy
    qz = cr * cp * sy - sr * sp * cy

    return qx, qy, qz, qw

def rpy_from_quat(x, y, z, w):

    srcp = 2*(w*x + y*z)
    crcp = 1-2*(x*x + y*y)
    roll = math.atan2(srcp, crcp)

    sp = 2*(w*y - z*x)
    if math.fabs(sp) >= 1:
        pitch = (sp/math.fabs(sp))*math.pi/2
    else:
        pitch = math.asin(sp)

```

```

    sycp = 2*(w*z + x*y)
    cycp = 1 - 2*(y*y + z*z)
    yaw = math.atan2(sycp, cycp)

    return roll, pitch, yaw

def main(args=None):

    #initial ROS2
    rclpy.init(args=args)

    #initial turtle client
    cli_obj = TurtleClient()
    cli_obj.get_logger().info('Turtlebot Client Started!')

    while rclpy.ok():

        cli_obj.update()
        rclpy.spin_once(cli_obj)

        unit_x = 1 #<put a reasonable ratio, 1 is a good number, around 1
is good enough>
        unit_z = 1 #<put a reasonable ratio, 1 is a good number, around 1
is good enough>

        ##### publish twist #####
        #cmd_msg = Twist()
        #cmd_msg.linear.x = float(50 * unit_x)
        #cmd_msg.angular.z = float(1 * unit_z)
        #cli_obj.twist_pub.publish(cmd_msg)

    # Destroy the node explicitly
    cli_obj.destroy_node()
    rclpy.shutdown()

if __name__ == '__main__':
    main()

```

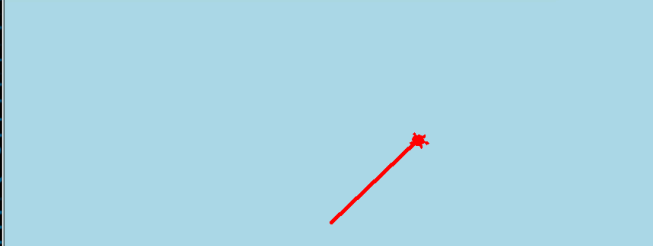


```
yahboom@yahboom-virtual-machine: ~/Desktop/Lab3/resou...
at run time.
If the overridden package is used by another package in any underlay, then the overriding package in the overlay must be API and ABI compatible or undefined behavior at run time may occur.

If you understand the risks and want to override a package anyways, add the following to the command line:
--allow-overriding teleop_twist_keyboard

This may be promoted to an error in a future release of colcon-overrides-check.
Starting >>> teleop_twist_keyboard
Finished <<< teleop_twist_keyboard [1.71s]

Summary: 1 package finished [2.11s]
yahboom@yahboom-virtual-machine: ~/Desktop/Lab3/resources_ws$ source install/setup.bash
yahboom@yahboom-virtual-machine: ~/Desktop/Lab3/resources_ws$ ros2 run python_turtle_turtlebot_server
[INFO] [1773405315.265793950] [turtleServer]: Turtlebot server started!
[INFO] [1773405336.87537575] [turtleServer]: Turtle color set: black
[INFO] [1773406269.958923922] [turtleServer]: Turtle color set: red
[INFO] [1773407143.349525549] [turtleServer]: Turtle color set: red
[INFO] [1773407649.300387865] [turtleServer]: Turtle color set: red
```



```
KeybaordInterrupt
[ros2run]: Interrupt
yahboom@yahboom-virtual-machine: ~/Desktop/Lab3/resources_ws$ ros2 run python_turtle_turtlebot_client --ros-args --param turtlecolor:=red turtlewidth:=5
Traceback (most recent call last):
  File "/home/yahboom/Desktop/Lab3/resources_ws/install/python_turtle/lib/python_turtle/turtlebot_client", line 33, in <module>
    sys.exit(load_entry_point('python-turtle==0.0.0', 'console_scripts', 'turtlebot_client'))()
  File "/home/yahboom/Desktop/Lab3/resources_ws/install/python_turtle/lib/python3.10/site-packages/python_turtle/turtlebot_client.py", line 104, in main
    rclpy.init(args=args)
  File "/opt/ros/humble/local/lib/python3.10/dist-packages/rclpy/__init__.py", line 89, in init
    return context.init(args, domain_id=domain_id)
```

```
For Holonomic mode (strafing), hold down the shift key:
-----
U      I      O
J      K      L
M      <      >

t : up (+z)
b : down (-z)

anything else : stop

q/z : increase/decrease max speeds by 10%
w/x : increase/decrease only linear speed by 10%
e/c : increase/decrease only angular speed by 10%

CTRL-C to quit

currently: speed 40.09 turn 1.00
currently: speed 44.10 turn 1.00
currently: speed 48.51 turn 1.00
currently: speed 53.36 turn 1.00
```

Task 3

```
yahboom@yahboom-virtual-machine: ~/Desktop/Lab3/resou...
yahboom@yahboom-virtual-machine: ~/Desktop/Lab3/resources_ws$ colcon build --packages-select python_turtle
Starting >>> python_turtle
Finished <<< python_turtle [3.01s]

Summary: 1 package finished [3.94s]
yahboom@yahboom-virtual-machine: ~/Desktop/Lab3/resources_ws$ source install/setup.bash
yahboom@yahboom-virtual-machine: ~/Desktop/Lab3/resources_ws$ ros2 launch python_turtle_turtle_teleop_launch.py
[INFO] [launch]: All log files can be found below /home/yahboom/.ros/log/2026-03-13-22-47-14-585471-yahboom-virtual-machine-9576
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [turtlebot_server-1]: process started with pid [9577]
[INFO] [turtlebot_client-2]: process started with pid [9579]
[INFO] [gnome-terminal-3]: process started with pid [9581]
[INFO] [gnome-terminal-3]: process has finished cleanly [pid 9581]
[turtlebot_server-1] [INFO] [1773413235.952400630] [turtleServer]: Turtlebot server started!
[turtlebot_server-1] [INFO] [1773413236.658790377] [turtleServer]: Turtle color set: red
[turtlebot_client-2] [INFO] [1773413236.696812165] [turtleClient]: Turtlebot Client Started!
```

```
Terminal
Moving around:
u      i      o
j      k      l
m      <      >

For Holonomic mode (strafing), hold down the shift key:
-----
U      I      O
J      K      L
M      <      >

t : up (+z)
b : down (-z)

anything else : stop

q/z : increase/decrease max speeds by 10%
w/x : increase/decrease only linear speed by 10%
e/c : increase/decrease only angular speed by 10%

CTRL-C to quit

currently: speed 0.50 turn 1.00
```

Task 4

